

Lecture 32: Stokes' Theorem (cont.) review

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS$$

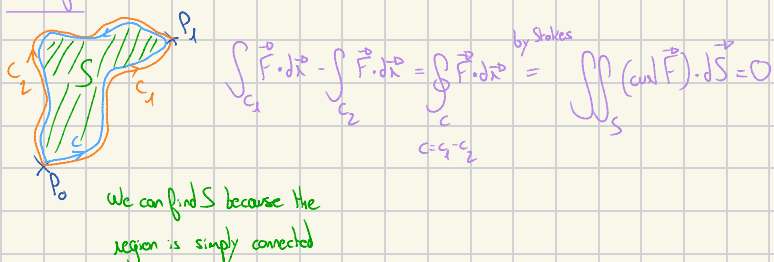
Stokes and path-independence: If we have a vector field, defined everywhere and it's $\text{curl} = 0$, then it's path-independent

Def: A region in space is simply connected if every closed loop inside that region bounds some surface inside this region

Recall: if $\vec{F} = \nabla f$ is a gradient field, then $\text{curl} \vec{F} = 0$

Theorem: if $\vec{F}$ is defined in a simply connected region and $\text{curl} \vec{F} = 0$ then $\vec{F}$ is a gradient field and $\oint_C \vec{F} \cdot d\vec{r}$ is path-independent, $\vec{F}$ is conservative etc.

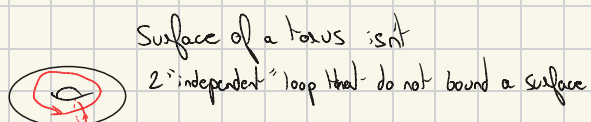
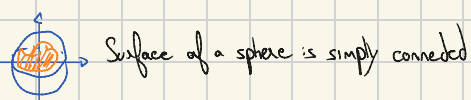
Proof: Assume $\text{curl} \vec{F} = 0$



$$\int_C \vec{F} \cdot d\vec{r} = \oint_C \vec{F} \cdot d\vec{r} = \oint_C \vec{F} \cdot d\vec{r} \stackrel{\text{by Stokes}}{=} \iint_S (\text{curl} \vec{F}) \cdot \hat{n} dS = 0$$

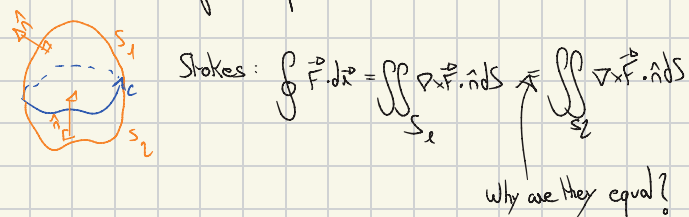
We can find S because the region is simply connected

Remark: Topology classifies surfaces in space.



Orientability: Ex: Mobius Strip
Non-orientable surface
Flux cannot be defined

Stokes and "surface independence"?



Stokes: $\oint_C \vec{F} \cdot d\vec{r} = \iint_{S_1} \nabla \times \vec{F} \cdot \hat{n} dS = \iint_{S_2} \nabla \times \vec{F} \cdot \hat{n} dS$
Why are they equal?

$$\iint_{S_1} \nabla \times \vec{F} \cdot \hat{n} dS - \iint_{S_2} \nabla \times \vec{F} \cdot \hat{n} dS = \iint_{S=S_1-S_2} \nabla \times \vec{F} \cdot \hat{n} dS \quad \text{by divergence thm (using the fact that } S \text{ is a closed surface)} = \iiint_D \text{div}(\nabla \times \vec{F}) dV$$

We can check that: $\text{div}(\nabla \times \vec{F}) = 0$ always

$$\vec{F} = \langle P, Q, R \rangle \quad \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \langle R_y - Q_z, R_z - P_x, Q_x - P_y \rangle \quad \text{div}(\nabla \times \vec{F}) = (R_y - Q_z)_x + (R_z - P_x)_y + (Q_x - P_y)_z = R_{yx} - Q_{zx} + P_{zy} - R_{zy} + Q_{xz} - P_{yz} = 0$$

So $\nabla \cdot (\nabla \times \vec{F})$ always 0

Divergence Thm

Stokes Thm

$$\iiint_D \text{div}(\vec{F}) dV = \iint_S \vec{F} \cdot \hat{n} dS \quad \text{Stokes Thm: } \iint_S (\nabla \times \vec{F}) \cdot \hat{n} dS = \oint_C \vec{F} \cdot d\vec{r}$$

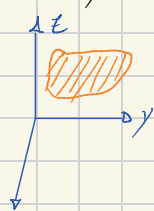
Rectangular Coord: $dV = dz dx dy$ or $dx dy dz$ etc.

Cylindrical: $dV = dz r dr d\theta$

Spherical: $dV = r^2 \sin \phi dr d\phi d\theta$

Formulas for $\hat{n} dS$. Ex: $\hat{n} dS = \dots dz dy$ becomes $\iint_D \dots$

* horizontal planes: yz -plane
 $dS = dy dz$ $\hat{n} = \pm \hat{i}$



Parametrize C, express in terms of a single variable

$$f(P_1) - f(P_0) = \int_C (\nabla f) \cdot d\vec{r}$$

Applications:

- * Mass
- * moment of inertia
- * Avg value of f
- * Gravitational attraction on mass at 0.

Spheres centered at 0

$$\hat{n} = \pm \frac{\langle x, y, z \rangle}{a}$$

$\uparrow$
radius

$$dS = a^2 \sin \phi d\phi d\theta$$

+ General Cases: $z = z(x, y)$ $\hat{n} dS = \langle -z_x, -z_y, 1 \rangle dx dy$

* $\hat{N}$ given: $\hat{n} dS = \pm \frac{\vec{N}}{|\vec{N}|} dx dy$

Cylinders around z -axis

$$\hat{n} = \pm \frac{\langle x, y, 0 \rangle}{a}$$

$$dS = a dz d\theta$$